ASPIRE FOR THE CLOUD: BUILDING APPLICATIONS WITH .NET ASPIRE

By Daniël Mertens

Clone the workshop repo: bit.ly/4cpjSZq

infoSupport
Solid Innovator

Daniël Mertens

- Info Support
  - Bouwen van software op maat
- Full stack .NET consultant
- Coding puzzles (Advent of Code, Hackatons, ...)
- Padel

Agenda

- Introduction. What’s Aspire?
- Demo
- Exercises
- Conclusions

What’s Aspire?

.NET Aspire provides tools, templates, and packages for building observable, production ready apps. .NET Aspire is delivered through a collection of NuGet packages that handle specific cloud-native concerns.

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What’s Aspire?

- Dev-Time Orchestration
- Integration
- Tooling
- Deployment

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Almost…

What’s Aspire?

Dev-Time Orchestration

Local dev experience matters

- How long does it take to setup your local dev environment?
  - High-friction onboarding = slow ramp-up time
  - Inconsistent dev experiences = hard to troubleshoot
  - Critical workflows rely on scripts, hacks, and tribal knowledge

Demo: Aspire Apphost

Dev-Time Orchestration

- Add the AppHost project.
- Start with a DistributedApplication

```
var builder = DistributedApplication.CreateBuilder(args);
```

- Add projects or other resources (unique naming)

```
builder.AddProject<Projects.GhostTracker_Bff>("bff");

builder.AddRabbitMQ("messaging").WithManagementPlugin()
```

- Run project

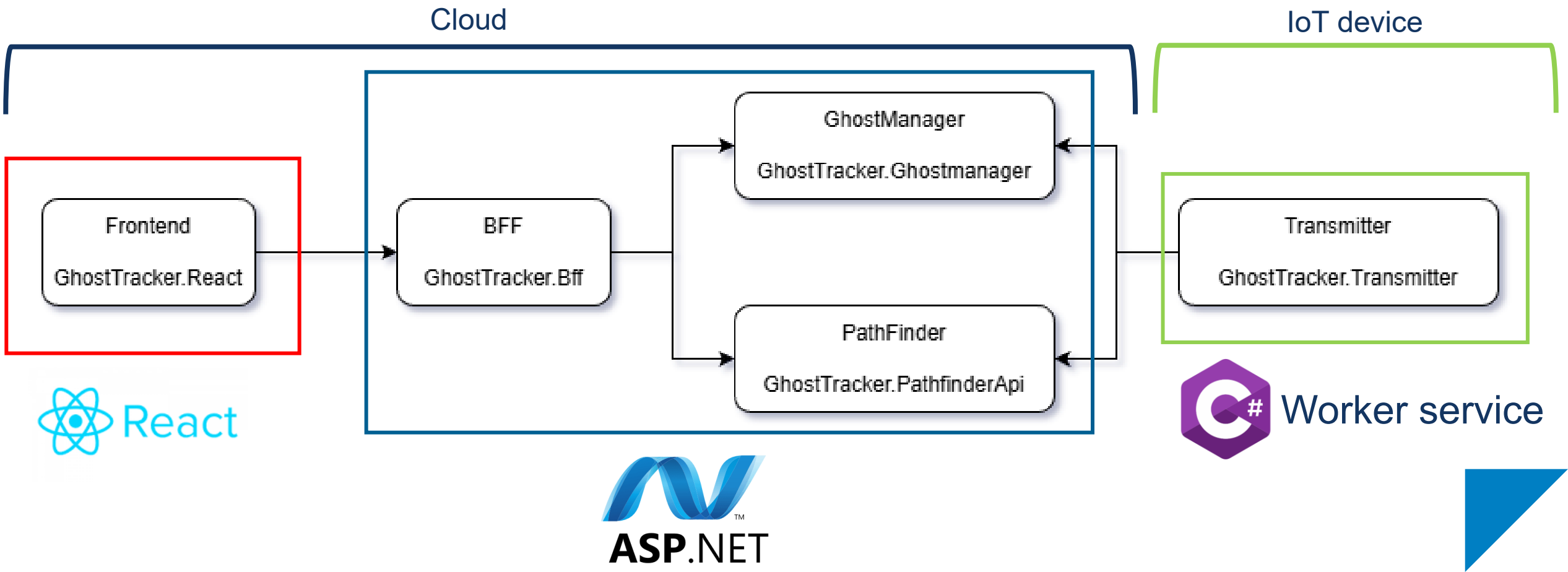
```
builder.Build().Run();
```

Tooling

- Everything in 1 convenient dashboard
- Overview of running projects
- Health checks
- Console output
- Environment variables

What are we going to build?

The ghost tracker application



Exercises

bit.ly/4cpjSZq

0. Environment setup
1. Project exploration and AppHost creation
2. Service orchestration
3. OpenTelemetry and observability
4. Service discovery
5. Running multiple instances
6. Message queue integration
7. Interaction service
8. Integration testing
9. Cloud deployment

Service defaults

- Telemetry
- Health checks
- Service discovery
- Resilience

Service defaults

- Default configuration as defined by Aspire
- Should be included in all projects.
- Name: <ProjectName>.ServiceDefaults

Exercises

0. Environment setup
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Integration Hosting

- Large collection of third party resources
- Nuget packages starting with “Aspire.Hosting”

Aspire.Hosting.Redis by aspire, Microsoft, **813K** downloads
Redis® support for .NET Aspire.

Aspire.Hosting.PostgreSQL by aspire, Microsoft, **706K** downloads
PostgreSQL® support for .NET Aspire.

Aspire.Hosting.SqlServer by aspire, Microsoft, **754K** downloads
Microsoft SQL Server support for .NET Aspire.

Aspire.Hosting.Azure.Storage by aspire, Microsoft, **547K** downloads
Azure Storage resource types for .NET Aspire.

Aspire.Hosting.NodeJs by aspire, Microsoft, **713K** downloads
Node.js support for .NET Aspire.

Aspire.Hosting.RabbitMQ by aspire, Microsoft, **410K** downloads
RabbitMQ support for .NET Aspire.

Integration Client

- Nuget packages starting with “Aspire” (no ‘Hosting’)
- To be used together with their hosting counterparts.

Aspire.RabbitMQ.Client by aspire, Microsoft, **294K** downloads
A RabbitMQ client that integrates with Aspire, including health checks, logging, and telemetry.

Aspire.StackExchange.Redis by aspire, Microsoft, **1,01M** downloads
A generic Redis® client that integrates with Aspire, including health checks, logging, and telemetry.

Aspire.Microsoft.Data.SqlClient by aspire, Microsoft, **150K** downloads
A Microsoft SQL Server client that integrates with Aspire, including health checks, metrics and telemetry.

Aspire.Azure.Storage.Blobs by aspire, Microsoft, **554K** downloads
A client for Azure Blob Storage that integrates with Aspire, including health checks, logging and telemetry.

Aspire.Microsoft.EntityFrameworkCore.SqlServer by aspire, Microsoft, **89**
A Microsoft SQL Server provider for Entity Framework Core that integrates with Aspire, including connection pooling, health check, logging, and telemetry.

Aspire.StackExchange.Redis.DistributedCaching by aspire, Microsoft, **455K**
A Redis® implementation for IDistributedCache that integrates with Aspire, including health checks, logging, and telemetry.

Integration

- Service discovery
  - Reference projects based on orchestration name

```
// Apphost
var rmq = builder.AddRabbitMQ("messaging")
    .WithManagementPlugin();

var manager = builder.AddProject<Projects.GhostTracker_GhostManager>("ghostmanagerapi")
    .WithReference(rmq);
```

```
// Manager startup
builder.AddRabbitMQClient("messaging");
```

Tooling

- Telemetry logging in dashboard
- Traces in dashboard
- Interactions
- Integration testing

Exercises

0. Environment setup
1. Project exploration and AppHost creation
2. Service orchestration
3. OpenTelemetry and observability
4. Service discovery
5. Running multiple instances
6. Message queue integration (Optional)
7. Interaction service
8. Integration testing
9. Cloud deployment

Aspire - Conclusion

Dev-Time Orchestration

Integration

Tooling

Vakantie job bij Info Support

- Project bouwen voor een VZW
  - 101 Genomes
- 1 maand, juli of augustus

Bedankt om aan de workshop mee te werken. Tot de volgende keer!

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Solid Innovator